Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table

```
SELECT column_name, data_type
FROM Target.INFORMATION_SCHEMA.COLUMNS
where table_name = 'customers'
```

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

B. Get the time range between which the orders were placed.

```
SELECT min(time(order_purchase_timestamp)) first_order,
       max(time(order_purchase_timestamp)) last_order
      FROM `Target.orders`
```

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	first_order	last_order
1	00:00:00	23:59:59

C. Count the Cities & States of customers who ordered during the given period.

```
select count(distinct(customer_city)) city,
       count(distinct(customer_state)) state
      from `Target.customers`
```

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	city	state
1	4119	27

II. In-depth Exploration:

A. Is there a growing trend in the no. of orders placed over the past years?

```
Select format_date('%Y-%m', order_purchase_timestamp) as yr_month,
      count(order_id) as total_orders
from `Target.orders`
```

```
group by yr_month
order by yr_month
```

Row	yr_month	total_orders
1	2016-09	4
2	2016-10	324
3	2016-12	1
4	2017-01	800
5	2017-02	1780
6	2017-03	2682
7	2017-04	2404
8	2017-05	3700
9	2017-06	3245
10	2017-07	4026

In december 2016 we can see decline in total_orders but after that numbers in total_order are kept increasing. We can definitely see growth in total_orders placesd over the past year.

B. Can we see some kind of monthly seasonality in terms of the no. of orders being

placed?

```
select format_date('%B', order_purchase_timestamp) as monthly_orders,
       count(order_id) as total_orders
from `Target.orders`

group by monthly_orders
order by total_orders desc
```

Row	monthly_orders	total_orders
1	August	10843
2	May	10573
3	July	10318
4	March	9893
5	June	9412
6	April	9343
7	February	8508
8	January	8069
9	November	7544
10	December	5674

In the month of may , july and august the total_orders are higher as compare to rest of the months.

C. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

```
select case when extract(hour from order_purchase_timestamp) between 0 and 6 then
'dawn'
            when extract(hour from order_purchase_timestamp) between 7 and 12 then
'morning'
            when extract(hour from order_purchase_timestamp) between 13 and 18 then
'afternoon'
            when extract(hour from order_purchase_timestamp) between 19 and 23 then
'night'
            end as time_of_the_day,
            count(order_id) as total_orders
from Target.orders
group by time_of_the_day
order by total_orders desc
```

Row	time_of_the_day	total_orders
1	afternoon	38135
2	night	28331
3	morning	27733
4	dawn	5242

In Afternoon Brazilian customers mostly place their orders .

III. Evolution of E-commerce orders in the Brazil region:

A. Get the month on month no. of orders placed in each state.

```
select c.customer_state,
       format_date('%m', order_purchase_timestamp) as month,
       count(order_id) as total_orders
from Target.customers as c
inner join Target.orders as o
on c.customer_id = o.customer_id

group by 1,2
order by 1,2
```

Row	customer_state	month	total_orders
1	AC	01	8
2	AC	02	6
3	AC	03	4
4	AC	04	9
5	AC	05	10
6	AC	06	7
7	AC	07	9
8	AC	08	7
9	AC	09	5
10	AC	10	6

B. How are the customers distributed across all the states?

```
select customer_state,
       count(distinct(customer_id)) as total_customer
from Target.customers

group by 1
order by 2 desc
```

Row	customer_state	total_customer
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

IV. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

A. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
with t1 as
(select format_date('%Y', order_purchase_timestamp) as yr_dt,
       sum(p.payment_value) as cost

from `Target.payments` p
inner join `Target.orders` o
on p.order_id = o.order_id
where format_date('%Y', order_purchase_timestamp) in ('2017', '2018')
and format_date('%m', order_purchase_timestamp) between '01' and '08'
group by 1 )

select *,
       round(100* (lead(cost,1) over(order by yr_dt) - cost)/cost,1) as perc_cost
from t1
```

```
order by yr_dt
```

Row	yr_dt	cost	perc_cost
1	2017	3669022.120000...	137.0
2	2018	8694733.840000...	*null*

B. Calculate the Total & Average value of order price for each state.

```
select c.customer_state,
       sum(it.price) as total_price,
       round(sum(it.price)/ count(distinct it.order_id),2) as avg_order_price
from `Target.order_items ` it
inner join Target.orders o
on o.order_id = it.order_id

inner join Target.customers c
on c.customer_id = o.customer_id
group by 1
```

Row	customer_state	total_price	avg_order_price
1	SP	5202955.049999...	125.75
2	RJ	1824092.669999...	142.93
3	PR	683083.7600000...	136.67
4	RS	750304.0200000...	138.13
5	MG	1585308.030000...	137.33
6	GO	294591.9499999...	146.78
7	SC	520553.3399999...	144.12
8	PB	115268.0800000...	216.67
9	DF	302603.9399999...	142.4
10	PA	178947.81	184.48

C. Calculate the Total & Average value of order freight for each state.

```
select c.customer_state,
       sum(it.price) as total_price,
       round(sum(it.freight_value)/ count(distinct it.order_id),2) as
avg_freight_value
from `Target.order_items ` it
inner join Target.orders o
on o.order_id = it.order_id

inner join Target.customers c
on c.customer_id = o.customer_id
```

```
group by 1
```

Row	customer_state	total_price	avg_freight_value
1	SP	5202955.049999...	17.37
2	RJ	1824092.669999...	23.95
3	PR	683083.7600000...	23.58
4	RS	750304.0200000...	24.95
5	MG	1585308.030000...	23.46
6	GO	294591.9499999...	26.46
7	SC	520553.3399999...	24.82
8	PB	115268.0800000...	48.35
9	DF	302603.9399999...	23.82
10	PA	178947.81	39.9

V. Analysis based on sales, freight and delivery time.

A. Find the no. of days taken to deliver each order from the order's purchase date

as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery

date of an order. Do this in a single query.

```
select order_id,
       date_diff(order_delivered_customer_date, order_purchase_timestamp, day) as
day_deliver,
       date_diff(order_delivered_customer_date, order_estimated_delivery_date, day)as
diff_estimated_delivery
from `Target.orders`
where order_status = 'delivered'
order by order_delivered_customer_date desc
```

Row	order_id	day_deliver	diff_estimated_d...
1	7e708aed151d6a8601ce8f2eaa...	136	96
2	450cb96c63e1e5b49d34f223f6...	143	106
3	b2997e1d7061605e9285496c5...	63	49
4	a2b4be96b53022618030c17ed...	66	41
5	7d09831e67caa193da82cfea3b...	50	36
6	f23681a0fffdb8051c674707c7e...	68	46
7	1e7d25f611e794f9614dd3e10a...	50	29
8	4af2fb154881f350d8696f7f7a7...	59	38
9	1b3190b2dfa9d789e1f14c05b6...	208	188
10	4505acb3759da6b9c7d79a80d...	44	33

B. Find out the top 5 states with the highest & lowest average freight value.

```
(select c.customer_state,
       avg(it.freight_value) as avg_fv,
       'top' as type
from `Target.order_items ` it
inner join `Target.orders` o
on o.order_id = it.order_id

inner join `Target.customers` c
on c.customer_id = o.customer_id
group by 1
order by avg_fv desc
limit 5)

union all

(select c.customer_state,
       avg(it.freight_value) as avg_fv,
       'bottom' as type
from `Target.order_items ` it
inner join `Target.orders` o
on o.order_id = it.order_id

inner join `Target.customers` c
on c.customer_id = o.customer_id
group by 1
order by avg_fv
limit 5)
```

Row	customer_state	avg_fv	type
1	RR	42.98442307692...	top
2	PB	42.72380398671...	top
3	RO	41.06971223021...	top
4	AC	40.07336956521...	top
5	PI	39.14797047970...	top
6	SP	15.14727539041...	bottom
7	PR	20.53165156794...	bottom
8	MG	20.63016680630...	bottom
9	RJ	20.96092393168...	bottom
10	DF	21.04135494596...	bottom

C. Find out the top 5 states with the highest & lowest average delivery time.

```
with cte1 as (
  select customer_state,
         round(avg(date_diff(o.order_delivered_customer_date,
o.order_purchase_timestamp,day)),2) as avg_delivery_time
  from Target.orders as o
  inner join `Target.customers` as c
  on o.customer_id = c.customer_id
  where o.order_status = 'delivered'
  group by customer_state
),
     cte2 as (
      select customer_state,
             avg_delivery_time,
             row_number() over(order by avg_delivery_time desc) as top,
             row_number() over(order by avg_delivery_time ) as bottom
      from cte1
     )
select customer_state,
       avg_delivery_time,
       case when top <= 5 then 'top5'
            when bottom <= 5 then 'bottom5'
       end as remark
from cte2
where top <= 5 or bottom <= 5;
```

Row	customer_state	avg_delivery_time	remark
1	SP	8.3	bottom5
2	PR	11.53	bottom5
3	MG	11.54	bottom5
4	DF	12.51	bottom5
5	SC	14.48	bottom5
6	PA	23.32	top5
7	AL	24.04	top5
8	AM	25.99	top5
9	AP	26.73	top5
10	RR	28.98	top5

D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
with cte1 as (
  select customer_id,
         date_diff(order_estimated_delivery_date, order_delivered_customer_date,
day)as diff_delv
  from `Target.orders`
  where order_status = 'delivered'
)

select c.customer_state,
       round(avg(ct.diff_delv),2) as avg_diff_dlv
from cte1 as ct
inner join `Target.customers` as c
using (customer_id)
group by c.customer_state
order by avg_diff_dlv desc
limit 5;
```

Row	customer_state	avg_diff_dlv
1	AC	19.76
2	RO	19.13
3	AP	18.73
4	AM	18.61
5	RR	16.41

VI. Analysis based on the payments:

A. Find the month on month no. of orders placed using different payment types.

```
select format_date('%Y', order_purchase_timestamp) as yr_dt,
       sum(p.payment_value) as cost

from `Target.payments` p
inner join `Target.orders` o
on p.order_id = o.order_id
```

Row	payment_type	yr_mt_dt	total_orders
1	UPI	2016-10	63
2	UPI	2017-01	197
3	UPI	2017-02	398
4	UPI	2017-03	590
5	UPI	2017-04	496
6	UPI	2017-05	772
7	UPI	2017-06	707
8	UPI	2017-07	845
9	UPI	2017-08	938
10	UPI	2017-09	903

B. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
Select payment_installments,
      count(order_id) as total_orders
from `Target.payments`
where payment_installments >= 1
group by 1
```

Row	payment_installm...	total_orders
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	5	5239
6	6	3920
7	7	1626
8	8	4268
9	9	644
10	10	5328